

BALVEER SINGH

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PROFESSIONAL SUMMARY

- Over 10+ years of experience in IT industry involving software development, **implementation**, production support, team leading, project management by executing projects to desired expectations.
- Expert in designing of **SSIS packages** from various sources like MS Access, Oracle, Excel, Microsoft SQL server, Flat file, and XML.
- Excellent technical and analytical skills with clear understanding of design goals for modelling of OLTP (ER) and dimensional modeling for OLAP.
- Involved in all phases of **SDLC** (Software Development Life Cycle) from analysis, design, development, testing and maintenance with timely delivery on different Business applications.
- Developed **PL/SQL programs** as Database Triggers, Stored Procedures, Functions and Packages.
- Good experience and understanding of implementing data warehousing programs and Data Integration solutions on Snowflake Cloud, AWS, Informatica Intelligent Cloud Services (IICS) with multiple relational databases (MySQL, Oracle, Teradata, SQL server).
- Experience in migrating other databases to Snowflake and AWS S3 bucket for integrating data from multiple source systems.
- Good knowledge of Oracle performance features like Query Optimizer, Execution plans, indexes, and performance tuning with HINTS, Explain plan.
- Strong Data Warehousing **ETL** experience of using Informatica 10.2/9.1 Power Center Client tools - Mapping Designer, Repository manager, Workflow Manager/Monitor and Server tools. Partitioned large tables using range partition technique and worked on Oracle supplied packages like DBMS_SQL, DBMS_JOB and UTL_FILE.
- Created Shell Scripts for invoking SQL scripts and scheduled them using crontab.
- Strong communication with the project team throughout all the stages of design, analytical problem solving and highly organized.
- Expertise in Client-Server application development using Oracle **12c/11g/10g/9i/8i**, **PL/SQL**, **SQL*PLUS**, **TOAD**, Workflow Builder and SQL*LOADER with experience in **UNIX** and **Windows**.
- Good Knowledge of TriZetto **FACETS**, an integrated application which aids in quick reprocessing by reducing the claim errors and experience supporting Provider/Finance/Billing/Claims systems.
- Extensive experience in Database **Performance Tuning**- Database Development including effective use of Database objects, SQL Trace, Explain Plan, Different types of Optimizers, Hints, Indexes, Table Partitions, Sub Partitions, Materialized Views, Global Temporary tables, Capabilities of using Oracle Built-in Functions.

EDUCATION:

- Bachelors of Computer Science from Chandigarh, University Aug 2006 to Apr 2010
- Master's Degree of Data Analytics from Boston University, May 2011.

PROFESSIONAL EXPERIENCE:

Massachusetts Mutual Life Insurance Company (MassMutual), Boston, MA
March 2021- Current
Senior Data Engineer

Responsibilities:

- Performed data analysis with PL./SQL and data analysis tools, SQL analyzer and generate reports, build dashboards to provide insights and improvement enhancements to stakeholders, resulting in 45% reduction in data retrieval time and improved data accessibility for the organization.
- Created PL/SQL stored procedures and functions for a streaming process and load the downstream data into AWS S3 buckets for storage and backup on AWS.
- Involved in implementing one time migration of data from Oracle to Snowflake tables from internal stage using Snow SQL.
- Worked on creating and loading the external stage (AWS S3), internal stage to transform data in Snowflake using COPY command.
- Created complex snowsql scripts and procedures for branching and looping in snowflake cloud data warehouse.
- Design, develop oracle database schemas, tables, standard views, materialized views, synonyms, unique indexes, non-unique indexes, constraints, triggers, sequences, cursor for loops, stand-alone procedures, functions, packages, and other database objects.
- Participate in cross functional project meetings as a subject matter expert for gathering requirements, testing implementing ETL solutions.
- Worked with teams to understand business needs and translate those business needs into actionable reports in Tableau and saving time.
- Optimized multiple views/procedures to use hash, materialize and parallel hints resulting in faster query response time.
- Development and Performance tuning of SSIS packages using counters, error handling, event handling with check points for transferring data between **Oracle** (FACETS) and **SQL, MongoDB** databases.
- Extensively worked on developing a **Data mart** for the FACETS to collect all the Invoice payments and collaborated internally with other subledger reports.
- Performed the uploading and downloading flat files from UNIX server using **FTP**.
- Involved in redesigning with implementation of table Partitions and Partition Indexes to make **Database** Faster and easier to maintain.
- Experience working in an Agile Environment and **DevOps** tool sets such as SVN, Red Gate, Bitbucket, GIT and Bamboo CI/CD integrations.
- Worked on Splunk reports and analyzed the API calls to improve performance of the queries and capture data from front end, middle ware application.
- Worked and revamped data dictionaries to create a robust plan for redesigning the DEAL platform and develop consistency across domain.
- Worked on ETL process – creating triggers and functions for extracting and loading data to a web app using PostgreSQL.
- Built ETL solutions that ingest transactional and event data for a web app with 10000 active users.
- Worked on partitioning strategies for large tables (500 GB) and improved the query performance by 45%.
- Design table structures and define ETL strategy to build performant Data solutions that are reliable.
- Used Splunk Streaming to consolidate and clean transactional data, resulting in speed improvements of 32% in the production web app.
- Resolved ETL issues by improving their performance using PL/SQL best practices and increasing the company's specialized web app functionality tremendously.
- Used **Pragma Autonomous** Transaction to avoid mutating problem in database trigger.
- Lead offshore developers and worked closely with QA and data warehouse teams to identify issues/bugs and resolve them in timely manner.
- Used python APIs for extracting daily data from multiple healthcare files.
- Handled errors using **Exception Handling** extensively for the ease of debugging and displaying the error messages in the application.
- Extracted complex data from various source systems like Oracle, SQL Server and DB2 to load the data into Landing Zone and then by using Java copy command loaded into AWS-S3 Raw Bucket.

- To increase the performance, balanced the input files of count against large files and loaded into AWS-S3 Refine Bucket and used copy command to achieve the micro-batch load into the Amazon Redshift.
- Improved response and continuous **enhancements** to queries up to 40 percent by reviewing and improving indexes, partitioning tables, and fixing of production issues.
- Worked on fixing issues with python and shell scripts that run using script snap to temporarily fix issues or missing functionalities like unzipping files and managing ftp files.
- Data loads have been performed by using the SQL loader, Imports & UTL files based on file formats.

State Street Corporation, Boston, MA
July 2019 - Mar 2021
Azure Data Engineer/Spark Developer

Responsibilities:

- Developed Python-based Spark applications (PySpark) in a distributed environment to load many CSV files with different schemas into Hive ORC tables.
- Involved in data transformations, mapping, and cleansing, monitoring, debugging, performance tuning, and troubleshooting Hadoop clusters.
- Developed a python script to transfer data, REST API's and extract data from on-premises to AWS S3. Implemented Micro Services based Cloud Architecture using Spring Boot.
- Conducted statistical analysis on healthcare data using Python and various tools.
- data scientists and software engineers, to integrate ChatGPT Collaborated with the data science team for data preprocessing, feature engineering, and production deployment of machine learning algorithms.
- Improved access time by refactoring data models, query optimization, and implementing Redis store to support Snowflake.
- Ingested data into Azure services such as Azure Data Lake, Azure Storage, Azure SQL, and Azure DW, and processed the data in Azure Databricks.
- Working with Informatics 9.5.1 and Informatics 9.6.1 Big Data edition. Scheduling the jobs
- Handle issues, bugs and provide solutions with root cause analysis and involve Datastax for product support if needed
- Created various types of data visualizations using Python and Tableau
- Collaborated with cross-functional teams, including into various applications and platforms
- Implemented data quality checks and validation processes to ensure the integrity and reliability of input data used to train and fine-tune ChatGPT models.
- Projects involve training staff in data governance and data quality, and driving well defined "quick win" projects to prove the usefulness of Data Governance at the organization.
- Data Lake Architecture planning and entity merge design – Healthcare
- Application Upgrades - ETL and Data Quality Upgrade from 9.6.1 to 10.2.
- Engineered the release, integration, and customization of Airflow for a 25+ person analytics team .
- Developed resulting scripts and workflow using Apache Airflow and shell scripting to ensure daily execution in production.
- Install and configure Apache Airflow for S3 bucket and Snowflake data warehouse and created dags to run the Airflow.
- Actively involved in working with various Azure cloud services, including HDInsight, Databricks, Data Lake, Blob storage, Data Factory, Synapse, SQL DB, and SQL DWH.
- Utilized Azure Data Factory, SQL API, and Mongo DB API to integrate data from sources such as Mongo DB, MS SQL, and GCP cloud storage (Blob, Azure SQL DB).
- Involved in writing unit tests, worked along with DevOps team in installing libraries, Jenkins agents and product ionized ETL jobs and micro services.
- Design and implement integrations from Collibra to other systems thru Collibra connect.
- Created and implemented custom asset model in Collibra for governing BJC measures and reports.
- Demonstrated ability to modularize Dagster pipelines, making components reusable and scalable. Experienced in creating parameterized pipelines and solid libraries, allowing for easy adaptation and extension of existing workflows for new use cases and data sources.

Boston Children's Hospital, Boston, MA
Apr 2016 – June 2019
AWS Cloud Data Engineer

Responsibilities:

- Architect the data lake by cataloging the source data, analyzing entity relationships, and aligning the design as per performance, schedule & reporting requirements.
- Architect BI applications as Enterprise Solution.
- Design and Develop Hadoop ETL solutions to move data to the data lake using big data tools like Sqoop, Hive, Spark, HDFS, Talend etc.
- Design and Develop Spark code using Python programming language & Spark SQL for high-speed data processing to meet critical business requirements.
- Preferable experience in using Kafka, Spark, ECS, Docker.
- Experience with AWS cloud services: EC2, EMR, RDS, Redshift, S3.
- Design and Develop ETL Processes in AWS Glue to migrate Campaign data from external sources like S3, ORC/Parquet/Text Files into AWS Redshift.
- Configured ECS using the docker image for python libraries needed for a pipeline.
- Implement RDD/Datasets/Data frame transformations in Scala through Spark Context and Hive Context.
- Import python libraries into the transformation logic to implement core functionality.
- Wrote Spark-SQL and embedded the SQL in Python files to run in the Airflow Dag as a part of pipeline.
- Used NOSQL database Amazon dynamo DB to store data of reporting Application.
- Written AWS Lambda code in Python for nested Json files, converting and consolidating the kinesis stream.
- Strong database development skills in Store procedure, Query language, performance optimization in RDBMS (DB2) as well as Cassandra and Hadoop.
- Involved in Migrating Objects from Oracle, Hive and Redshift to Snowflake.
- Design and construct of AWS Data pipelines using various resources in AWS including AWS API Gateway to receives response from AWS Lambda and retrieve data from snowflake using lambda function and convert the response into Json format using Database as Snowflake, DynamoDB, AWS Lambda function and AWS S3.
- Constructing a state-of-the-art data lake on AWS using EMR, Spark, Kafka, Python.
- Partners with DBT on delivery of Data definitions and aligns with Instance Data conversion team.
- Develop algorithms & scripts in Hadoop to import data from source system and persist in HDFS (Hadoop Distributed File System) for staging purposes.

Verizon Wireless, Boston, MA
Oct 2014 – Mar 2016
Oracle Developer

Responsibilities:

- Monitored the Oracle SQL query performance using EXPLAIN PLAN.
- Wrote sequences for automatic generation of unique keys to support primary and foreign key constraints in data conversions.
- Generated server-side **PL/SQL** scripts and packages for data manipulation and enhance the performance.
- Schedule and maintain SSIS packages daily, weekly and monthly using SQL server Agent.
- Involved in maintaining batch processing **PL/SQL** procedures for data extraction for feed creation.
- Supported VMware, ESX vSphere 5 to monitor, optimize performance and resolve production issues.
- Used **DTS Packages** as ETL tool for migrating data from SQL server to Oracle 10g.
- Performed query analysis and enhanced Query performance with Oracle indexing features and table partitioning features.
- Involved in creating standardized Crystal Reports to support internal and external customers and running ad-hoc queries that provide the data necessary to measure business performance and analyze trends.

- Conducted database and code tuning to improve performance of the application, used Bulk binds, in-line queries, Dynamic SQL, Analytics and Sub-query factoring etc.
- Involved in the complete life cycle of the product from requirements gathering to end of system testing.
- Involved in analyzing business requirements and designing the functional specifications and translated functional requirements into technical specs.
- Experience in developing efficient **PL/SQL** code using stored procedure, functions and triggers.
- Collaborated with cross functional Business and Technology teams to define project requirements.
- Generated Reports using Global Variables, Expressions and Functions for the reports.
- Worked on high performance queries/programs using Collections, Bulk Binds, Objects, Nested tables, REF Cursors, pipeline functions etc.
- Used Oracle DBMS packages to carry out dynamic SQL processing.
- Worked on developing, deploying, and monitoring packages for **ETL** tasks using Data Transformation Services (DTS) Import/Export Utility, and Bulk Copy Program and DTS packages.
- Developed batch jobs, shell scripts to automate file transfers, data cleansing, loading and final data processing.
- Coordinated with DBA in creating and managing indexes, table spaces, and auditing and data quality checks.

YMCA of Metropolitan Washington, Washington DC

Aug 2011 – Sep 2014

PL/SQL Developer

Responsibilities:

- Developed stored procedures, functions, packages, and SQL Scripts using **PL/SQL**.
- Performed onsite support and monitoring of AS400 and two mainframes and upgrading network virtualization components.
- Developed procedures for efficient error handling process by capturing errors into user managed tables.
- Evaluated Database performance and perform maintenance duties such as tuning, backup, restoration of data.
- Code optimization and performance tuning of the application using Explain Plan, hints, indexing and partitioning of tables.
- Created support documents for the above projects in **Visio** like how the data flows from one table to another with clear job plans.
- Developed **PL/SQL** scripts, packages, procedures, cursors, tables, views, materialized views, and functions as per the business requirements.
- Experience in handling very large data files using **SQL*Loader** and external tables.
- Knowledge on VMware ESX host issues in different environments.
- Scheduled Reports on daily and weekly basis in Report manager and email them to managers and analysts in CSV format.
- Developed SQL and **PL/SQL** scripts to transfer Data across the schemas and databases.
- Created SSIS packages to perform filtering options and to import the data on daily basis from file server into database tables.
- Worked in SSIS to create **ETL** Packages to perform extract, transform and load data from different source to destination tables.